

STUMPRIDER · CALCULATION NOTE

StumpRider sizing calculations

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DOCUMENT CONTROL

Document ID	SMR-CAL-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First issue at TRL 3, on the constructable design of SMR-DDR-002

StumpRider sizing calculations

On paper, a 2.0 mm soft aluminium shear pin (302 N nominal) frees a net hooked over a branch with a margin of at least two, and breaks before the pull can heel the 8 m design canoe more than 4.4 deg. The tool reaches 4.18 m from the top hand to the ring and the tether works snags to 8 m. Four results fall short: a net wrapped once round a branch needs more pull than the pin allows (R1); the same pin would heel a small dugout about 24 deg (R3); the whole kit weighs 6.5 kg (R7); and a kit costs USD 92.50 at prototype prices (R10). Each is posed as a decision for Amish in `docs/REVIEW.md`.

Every figure comes from `docs/04-calcs/sizing.py`, which imports the parametric model (`cad/src/model.py`), so the sizes here are those of the STEP files, the drawings and the build plan. Tags in square brackets match the script output and `docs/04-calcs/results.csv`. These are screening estimates for a paper proof of concept; they do not replace the pin calibration on CalRig, the canoe heel test or the staged-snag trials, which are TRL 4 work.

Safety: The pin rating is a safety limit. It is valid only for pins cut from a reel whose sample has been broken on CalRig within 256 to 347 N, and only for canoes like the design canoe. Until a heel test says otherwise the tool is not used from canoes under 7 m or with fewer than three crew, the operator kneels, and every person wears a life jacket.

■ 1. Assumptions

Table 1. Assumptions.

#	Assumption	Value	Basis
A1	Shear strength of the pin wire	48 MPa	1050 or 1100 aluminium, O temper: about 0.6 of an 80 MPa tensile strength; to be measured per reel
A2	Design canoe	8.0 m long, 1.10 m waterline beam, 0.60 m deep, hull 300 kg, three crew of 65 kg, 55 kg of net and gear (550 kg)	A paddled Volta plank canoe; to be replaced by LevelHull or partner measurements
A3	Small dugout (hard case)	5.5 m, 0.75 m beam, 0.45 m deep, hull 120 kg, two crew, 30 kg load (280 kg)	Smallest canoe likely to carry the tool
A4	Canoe hydrostatics	Waterplane transverse moment $0.05 L B^3$; block coefficient 0.45; centre of buoyancy at 0.55 of the draft; hull mass centre 0.22 to 0.25 m up; kneeling crew 0.50 to 0.55 m up	Screening values for narrow round-bilged hulls
A5	Working posture	Pole 20 deg off vertical; operator's hands 0.35 m (0.25 m in the dugout) off the centreline and 0.30 m (kneeling) or 0.85 m (standing) above the gunwale	The crutch is clamped near the bow
A6	Net line forces	Crew holds 100 N while the ring is run down and slackens to 30 N while the operator lifts; the far leg of the net pulls 20 N	Operating rule and the net's own drag
A7	Friction of wet netting on wet bark	0.3 to 0.5	Range for nylon on rough wood
A8	Materials	Steel 7,850 kg/m ³ , S235; 6063-T6 aluminium 2,700 kg/m ³ , E 69 GPa, bearing 320 MPa; stainless clevis pins 300 MPa in shear; 4.8 mm stainless rivet 2.9 kN; HDPE 950; foam 30 kg/m ³ ; cord 25 g/m	Handbook and catalogue values
A9	Heel limit	5 deg with the pin at the top of its band	Conservative: well short of the 40 deg at which the design canoe's gunwale goes under, and small enough for kneeling crew to keep their balance

■ 2. Shear pin

The pin passes through both fork cheeks and the tang, so it is cut on two planes. Rating = 2 x shear strength x pin area = 2 x 48 MPa x 3.14 mm² = 302 N nominal [P1]. R3 allows plus or minus 15 %: 256 N [P2] to 347 N [P3]. Every safety check below uses the top of the band; every freeing check uses the bottom.

■ 3. Canoe heel

When the operator pulls up on the pole, the pole pulls down and outboard on the operator's hands, and so on the canoe. The heeling moment is the pull times (cos 20 deg x hands off centreline + sin 20 deg x hands above the water), plus the crew's slack line over the crutch. The righting moment is displacement x g x GM x sin(heel).

Table 2. Heel when the pin breaks at the top of its band.

Canoe	Displacement [H1, K1]	GM [H2, K2]	Freeboard [H3, K3]	Gunwale under at [H4, K4]	Heel, kneeling [H5, K5]	Heel, standing [H6, K6]	Pull for 5 deg, kneeling [H7, K7]
Design canoe	550 kg	0.69 m	461 mm	40 deg	3.4 deg	4.4 deg	523 N
Small dugout	280 kg	0.15 m	299 mm	39 deg	23.7 deg	34.1 deg	53 N

The pin protects the design canoe with a margin of 1.5 over the top of its band (kneeling). It does not protect the small dugout: 53 N heels it 5 deg, less than half of what a hooked net needs. A lighter 1.4 mm pin (148 N nominal [O11]) still heels the dugout 12.2 deg [L5]. Small canoes are therefore not covered by this design; the interim rule (no use from canoes under 7 m or with fewer than three crew) stands until a heel test, and the long-term answer is a decision for Amish (R3).

■ 4. Pull needed to free the net

The net lies over the branch with a contact angle θ . To lift it, the ring must overcome the far leg's tension multiplied by the capstan factor $e^{(\mu \theta)}$, plus the crew's line: pull = $20 \text{ N} \times e^{(\mu \theta)} + 30 \text{ N}$.

Table 3. Pull to lift the net off the branch.

Snag	Friction 0.3	Friction 0.5	Against the pin band (256 to 347 N)
Hooked, half a turn	81 N [F1]	126 N [F2]	Frees, margin 2.0 at the bottom of the band [F5]
Wrapped once round, one and a half turns	368 N [F3]	2,260 N [F4]	Breaks the pin first

If the crew keeps full tension while the operator lifts, the hooked case rises to 196 N [F6], still inside the band; slackening the line is the operating rule. A wrapped net cannot be lifted straight off: the ring has to walk it back round the branch half a turn at a time (each half turn divides the pull by 1.6 to 4.8), or the net is cut. R1 is at risk until the staged-snag trials show how often nets wrap and whether the ring can unwind them.

■ 5. Reach

- Top hand to ring centre: 4.18 m [R1] (three 1.45 m sections, hands 0.3 m below the cap). R2 met.
- Depth reached at 20 deg lean, kneeling: 3.17 m below the surface [R2].
- Tether cord: 10 m, with 2 m kept above the water to the hands: 8 m below the surface [R3]. R2 met.
- Ring inside diameter 100 mm [R4]; the hinged ring opens fully, so any line from 4 to 16 mm goes in. R6 met.

■ 6. Strength and stiffness

Table 4. Strength checks against the top of the pin band (347 N).

Check	Result
Euler buckling of the whole pole, 4.35 m, pinned ends	767 N [S1]; factor 2.2 [S2]
Lock pins, 6 mm stainless, double shear	factor 49 [S3]
Pole wall bearing at a lock pin hole	factor 22 [S4]
Sleeve rivets, two per sleeve (pull only; push butts the tubes)	factor 17 [S5]
Tang tear-out above the shear pin hole	factor 44 [S6]
Ring bending, pulled at the tang against the far side	36 MPa [S7]; factor 6.5 on yield [S8]
Tang bending, a quarter of the pull acting sideways	factor 6.5 [S9]
Crutch axle, M12, 500 N line tension over the roller	92 MPa [S10], below the 240 MPa of grade 4.6

Everything except the pin is at least twice as strong as the pin, so the pin is always the part that lets go. The buckling factor of 2.2 is the smallest margin; the sleeves stiffen the joints, which the Euler figure ignores.

■ 7. Float

The float winder gives 1.26 kg of net lift [B1]. The ring and the cord weigh 0.60 kg in water [B2], so a dropped ring and cord float with a reserve of 2.1 [B3]. With the jiggling weight on, the load is 1.57 kg [B4], more than the float holds: in jiggling use the cord's end is tied to a thwart.

■ 8. Mass

Table 5. Masses from the model.

Item	Mass
Rider ring with hinge bolt and gate pin	0.61 kg [M1]
Pole head	0.56 kg [M2]
Pole sections, sleeves, rivets, lock pins, cap	2.56 kg [M3]
Jiggling weight	1.11 kg [M4]
Gunwale crutch	1.37 kg [M5]
Tether cord, float winder, spare pins	0.29 kg [M6]
Working tool (ring, head, pole, pins)	3.73 kg [M7]
Whole kit	6.5 kg [M8]

The longest packed piece is a section with its sleeve, 1.55 m [M9]; packed length meets R7. The kit mass does not: the jiggling weight and the steel crutch are 2.5 kg of it. Options (decision for Amish): crutch left clamped on the canoe and weight kept at the landing, 4.02 kg carried [O9]; the same with bamboo sections, 3.37 kg [O10]; weight left out only, 5.39 kg [O1]; weight out and a hardwood crutch block, 4.9 kg [O4].

■ 9. Build time and cost

- Workshop time for one kit, one smith: 7.7 h [T1] (estimate, galvanising sent out). R9 met on paper, with little to spare.
- Parts cost of one kit at prototype prices: USD 92.50 [C1]. R10 not met. The pole, sleeves and rivets are 31 % [C5], galvanising 13 % [C6] and the crutch 13 % [C7].
- Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 284 for the prototype run of three kits and calibration wire (USD 1,716 under the target) [C2 to C4].
- Options for R10 (decision for Amish): bamboo pole sections with steel ferrules, USD 70.50 [O5]; zinc-rich paint in place of galvanising, USD 83.50 [O6]; crutch and weight shared, one set per five canoes, USD 74.90 [O7]; bamboo and sharing together, USD 52.90 [O8].

■ 10. Results against the requirements

Table 6. Summary.

Requirement	Result	Status
R1	Hooked nets 81 to 126 N; wrapped nets 368 N to 2.3 kN	At risk
R2	4.18 m top hand to ring; tether to 8 m	Met on paper
R3	Design canoe 3.4 to 4.4 deg; small dugout 24 deg; band depends on calibration	At risk
R4	Pull capped at 347 N; mesh damage not calculable	Cannot be shown on paper
R5	Release time	Cannot be shown on paper
R6	100 mm hinged ring	Met on paper
R7	Packed 1.55 m; mass 6.5 kg	Not met (mass)
R8	Galvanised steel, aluminium, stainless, polymers	Met on paper
R9	7.7 h	Met on paper
R10	USD 92.50 per kit	Not met